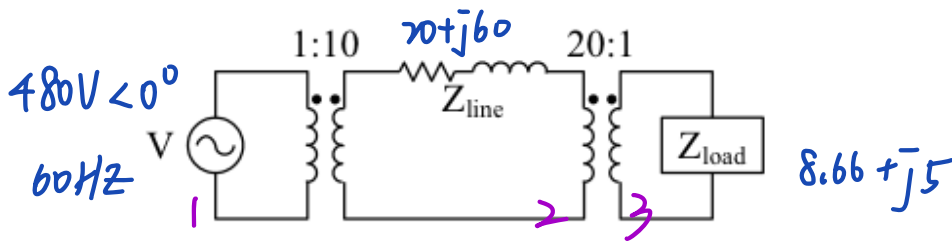


0419

From CH2 Transformers Ex 2-3

5.(20) A simple power system is shown in figure. This system contains a $480\text{V}(0^\circ)/60\text{Hz}$ generator connected to an ideal $1:10$ step-up transformer, a transmission line, an ideal $20:1$ step-down transformer, and a load. The impedance of the transmission line is $Z_{\text{line}} = 20 + j60\Omega$, and $Z_{\text{load}} = 8.66 + j5\Omega$. The base values for this system are chosen to be 480V and 10kVA at the generator.

- Find the base voltage, current, impedance, and apparent power at every point in the power system.
- Convert this system to its per-unit equivalent circuit.
- Find the power supplied to the load in this system.
- Find the power lost in the transmission line



<math>\angle 0^\circ</math>

1a) $V_{\text{base1}} = 480\text{V}$, $S_{\text{base1}} = 10\text{kVA}$

$$I_{\text{base1}} = \frac{S_1}{V_1} = \frac{10000}{480} = 20.83\text{A}$$

$$Z_{\text{base1}} = \frac{V_1}{I_1} = \frac{480}{20.83} = 23.04\Omega$$

$$V_{\text{base2}} = 10 \cdot V_{\text{base1}} = 4800\text{V}, S_{\text{base2}} \text{ 不变 } = 10\text{kVA}$$

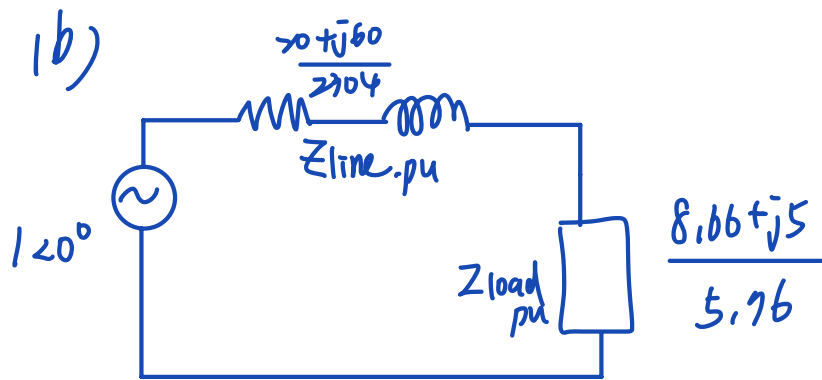
$$I_{\text{base2}} = \frac{S_2}{V_2} = \frac{10000}{4800} = 2.083\text{A}$$

$$Z_{\text{base2}} = \frac{V_2}{I_2} = \frac{4800}{2.083} = 2304\Omega$$

$$V_{\text{base3}} = V_{\text{base2}} \times \frac{1}{20} = 240\text{V}, S \text{ 不变 } = 10\text{kVA}$$

$$I_{\text{base3}} = \frac{S_3}{V_3} = \frac{10000}{240} = 41.67\text{A}$$

$$Z_{\text{base3}} = \frac{V_3}{I_3} = \frac{240}{41.67} = 5.76\Omega$$



• generator

$$V_{G.pu} = \frac{V_G}{V_{base1}} = \frac{480\angle 0^\circ}{480V} = 1\angle 0^\circ pu$$

$$Z_{line.pu} = \frac{20 + j60}{2704} = 0.0087 + j0.026 pu$$

$$Z_{load.pu} = \frac{8.66 + j5}{5.76} = (1.503 + j0.868) pu$$

(c)

$$I_{pu} = \frac{V_{pu}}{Z_{tot.pu}} = \frac{1\angle 0^\circ}{(0.0087 + j0.026) + (1.503 + j0.868)} = 0.569\angle -30.6^\circ pu$$

$$P_{load.pu} = I_{pu}^2 \cdot R_{load.pu} = (0.569)^2 (1.503) = 0.487$$

$$P_{load} = P_{load.pu} \cdot S_{base} = 0.487 \times 10^4 VA = 4870 W$$

(d)

$$P_{line.pu} = I_{pu}^2 \cdot R_{line.pu} = (0.569)^2 (0.0087) = 0.00282$$

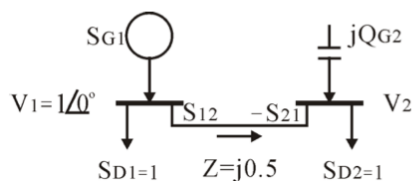
$$P_{line} = P_{line.pu} \cdot S = 0.00282 \cdot 10^4 = 28.2 W$$

* 0419. Ch2 EX 2.14

(第21回)

problem 6.

6.(20) $S_{G1}: V_1 = 1 \angle 0^\circ$, $S_{D1} = 1 + jQ_{G2}$, $V_2 = ?$, $S_{D2} = 1$, $Z = j0.5$, (a) Find Q_{G2} for $|V_2| = 1$ (b) and $\angle V_2$? (c) If $Q_{G2} = 0$, could be supplied load S_{D2} ? (d) and $\angle V_2$?



<sol> (a) & (b)

$$\because S_{D2} = 1 \in \mathbb{R}$$

Q_{G2} 純虚数

$$\therefore P_{12} = -P_{21} = 1$$

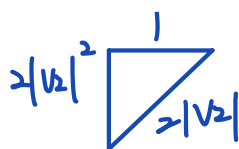
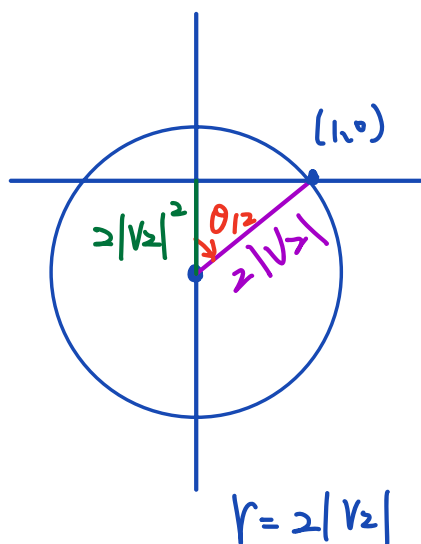
$$P_{12} = \frac{|V_1||V_2|}{X} \sin \theta_{12} = \frac{1 \cdot 1}{0.5} \sin \theta_{12} = 1$$

$$\therefore \sin \theta_{12} = \frac{1}{2} \rightarrow \theta_{12} = 30^\circ \text{ and } \angle V_2 = -30^\circ$$

$$Q_{G2} = Q_{21} = \frac{|V_2|^2}{X} - \frac{|V_2||V_1|}{X} \cos \theta_{12}$$

$$= \frac{1^2}{0.5} - \frac{1 \cdot 1}{0.5} \cdot \frac{\sqrt{3}}{2} = 2 - \sqrt{3} = 0.268$$

(c) if $Q_{G2} = 0$, $-S_{21} = S_{D2} = 1$



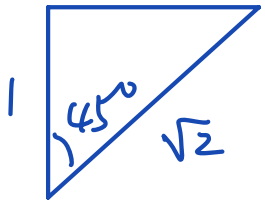
$$\Rightarrow 1^2 + (2|V_2|^2)^2 = (2|V_2|)^2$$

$$\Rightarrow 4|V_2|^4 - 4|V_2|^2 + 1 = 0$$

$$x = |V_2|^2, \quad 4x^2 - 4x + 1 = 0$$

$$x = \frac{1}{2}, \therefore |V_2| = \frac{1}{\sqrt{2}}$$

$$V_2 = \frac{1}{\sqrt{2}}$$

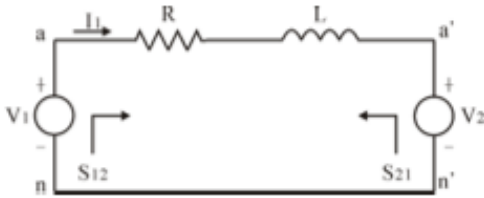


∴ 可以用 $V_2 = \frac{1}{\sqrt{2}} \angle -45^\circ$ 來供應負載

★ Problem 7.

7. (20) Find (a) S_{12} and S_{21} ; (b) P_{12} and $-P_{21}$; (c) Q_{12} and Q_{21} .

($Z = R + j\omega L$, $V_1 = |V_1|e^{j\theta_1}$, $V_2 = |V_2|e^{j\theta_2}$, $Z = |Z|e^{j\angle Z}$, $\theta_{12} = \theta_1 - \theta_2$)



sol >.

$$S_{12} = V_1 I_1^* = V_1 \left(\frac{V_1 - V_2}{Z} \right)^* = \frac{|V_1|^2}{Z^*} - \frac{V_1 V_2^*}{Z^*}$$

$$= \frac{|V_1|^2}{|Z|} e^{j\angle Z} - \frac{|V_1||V_2|}{|Z|} e^{j\angle \theta_1 - \theta_2 + \angle Z}$$

$$= \frac{|V_1|^2 e^{j\angle Z}}{|Z|} - \frac{|V_1||V_2|}{|Z|} e^{j\angle Z} e^{j\theta_{12}}$$

$$S_{21} = V_2 (-I_1)^*$$

$$= \frac{|V_2|^2}{|Z|} e^{j\angle Z} - \frac{|V_2||V_1|}{|Z|} e^{j\angle Z} e^{j\theta_{12}} \quad \#$$

$$10) \begin{cases} S_{12} = C_1 - B e^{j\theta_{12}} \\ -S_{21} = C_2 + B e^{-j\theta_{12}} \end{cases}$$

$$C_1 = \frac{|V_1|^2}{|Z|} e^{j\angle Z}, \quad C_2 = \frac{-|V_2|^2}{|Z|} e^{j\angle Z}, \quad B = \frac{|V_1||V_2|}{|Z|} e^{j\angle Z}$$

$\downarrow$ 送端圓圓心 $\downarrow$ 受端圓圓心 $\downarrow$ 半徑

Simply, assume $R=0$, $Z = R + j\omega L = j\omega L = jX$

$$|Z| = X, \quad \angle Z = 90^\circ$$

代入 $S_{12} \cdot S_{21}$

$$\begin{aligned} S_{12} &= \frac{|V_1|^2}{|Z|} (\cos 90^\circ + j \sin 90^\circ) - \frac{|V_1||V_2|}{|Z|} \cdot (\cos 90^\circ + j \sin 90^\circ) (\cos \theta_{12} + j \sin \theta_{12}) \\ &= j \frac{|V_1|^2}{X} - j \frac{|V_1||V_2|}{X} \cos \theta_{12} + \frac{|V_1||V_2|}{X} \sin \theta_{12} = P_{12} + jQ_{12} \end{aligned}$$

$$\therefore P_{12} = \frac{|V_1||V_2|}{X} \sin \theta_{12}$$

$$Q_{12} = \frac{|V_1|^2}{X} - \frac{|V_1||V_2|}{X} \cos \theta_{12}$$

$$S_{21} = V_2 (-I_1)^*$$

$$= \frac{|V_2|^2}{|Z|} e^{j\angle Z} - \frac{|V_2||V_1|}{|Z|} e^{j\angle Z} e^{-j\theta_{12}}$$

$$\angle Z = 90^\circ, |Z| = X$$

$$\rightarrow \frac{|V_2|^2}{X} (\cos 90^\circ + j \sin 90^\circ) - \frac{|V_2||V_1|}{X} (\cos 90^\circ + j \sin 90^\circ) (\cos \theta_{12} - j \sin \theta_{12})$$

$$= j \frac{|V_2|^2}{X} - j \frac{|V_2||V_1|}{X} \cos \theta_{12} - \frac{|V_2||V_1|}{X} \sin \theta_{12} = P_{21} + jQ_{21}$$

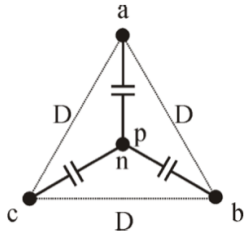
$$\therefore P_{21} = \frac{-|V_2||V_1|}{X} \sin \theta_{12} = -P_{12}$$

$$Q_{21} = \frac{|V_2|^2}{X} - \frac{|V_2||V_1|}{X} \cos \theta_{12}$$

#

0531.

1. (20) Assume that 1. conductors are equally spaced, D , and have equal radii r . 2. $q_a + q_b + q_c = 0$ ($c_a = c_b = c_c = c$, $v_a + v_b + v_c = 0$) 3. $i_a + i_b + i_c = 0$. Find (a) $c = ?$ (b) $\lambda_a = ?$



<sol>.

(a.)

$$V_a = \frac{1}{2\pi\epsilon} (q_a \ln \frac{1}{r} + q_b \ln \frac{1}{D} + q_c \ln \frac{1}{D}) = \frac{1}{2\pi\epsilon} (q_a \ln \frac{1}{r} - q_a \ln \frac{1}{D})$$

$$= \frac{1}{2\pi\epsilon} \cdot q_a \left(\ln \frac{D}{r} \right)$$

$$C = \frac{q}{V}, c_a = c_b = c_c = c = \frac{q_a}{\frac{1}{2\pi\epsilon} q_a \ln \frac{D}{r}} = \frac{2\pi\epsilon}{\ln \frac{D}{r}} \quad (F/m)$$

(b)

$$\lambda_a = ?$$

$$\lambda_a = \frac{\mu_0}{2\pi} \left\{ \bar{i}_a \cdot \left(\frac{\mu_r}{4} + \ln \frac{1}{r} \right) + \bar{i}_b \ln \frac{1}{D} + \bar{i}_c \ln \frac{1}{D} \right\}$$

$$= \frac{\mu_0}{2\pi} \left\{ \bar{i}_a \cdot \left(\frac{\mu_r}{4} + \ln \frac{1}{r} \right) - \bar{i}_a \ln \frac{1}{D} \right\}$$

$$= \frac{\mu_0 \bar{i}_a}{2\pi} \left\{ \ln \left(\frac{1}{r} \cdot e^{\frac{\mu_r}{4}} \right) - \ln \frac{1}{D} \right\}$$

$$= \frac{\mu_0}{2\pi} \bar{i}_a \left\{ \ln \frac{1}{r'} - \ln \frac{1}{D} \right\} = \frac{\mu_0}{2\pi} \bar{i}_a \ln \frac{D}{r'}, \quad \frac{1}{r'} = \frac{e^{\frac{\mu_r}{4}}}{r}$$